

ADITYA

Aerodynamics & CFD Engineer | MSc Advanced Aeronautical Engineering, Imperial College London

London, UK | +44 7549 349926 | aditya.pravega@gmail.com | linkedin.com/in/aditya-kumar-747bcr7 | adityapravega-create.github.io/portfolio

PROFILE

Aerodynamics and CFD engineer with an MSc in Advanced Aeronautical Engineering from Imperial College London, completed September 2026, and four years developing aerodynamic packages in Formula Student across two teams. My core work is external aerodynamic design validated by CFD — wings, diffusers, undertrays and bodywork in STAR-CCM+ and Ansys Fluent — together with the internal flow and thermal cases that come with them. My MSc thesis went down to the equations, deriving the RANS equations in a moving reference frame and extending a Python Euler solver to rotation as an independent check on the commercial result. What I care about most is correlation: I ran the coast-down and tuft tests on the car myself and correlated them against the CFD, and I follow designs through to hardware in CAD, FEA, GD&T drawings, mould design and composite layup. Available immediately.

EDUCATION

Imperial College London

Sep 2025 – Sep 2026

MSc Advanced Aeronautical Engineering

London, UK

- Modules: Turbulence and Turbulence Modelling, Applied Aerodynamics, Flow Instability and Transition, Finite Elements, Lightweight Structures, Design Optimisation, Sustainable Aviation.

Vellore Institute of Technology

Sep 2021 – Jul 2025

B.Tech Mechanical Engineering (Automotive Specialisation), CGPA 8.66 / 10

Vellore, India

- Modules: Computational Fluid Dynamics, Fluid Dynamics, Vehicle Body Aerodynamics, Vehicle Dynamics, Thermal Engineering, Internal Combustion Engines, Design of Machine Elements, Solid Mechanics.

TECHNICAL SKILLS

CFD & Simulation: STAR-CCM+, Ansys Fluent, OpenFOAM, XFLR5; steady and transient RANS, turbulence modelling (k-omega SST, k-epsilon realizable, Spalart-Allmaras), moving reference frame and rotating domains, conjugate heat transfer, multiphase VOF; mesh generation, mesh independence and domain sizing studies, convergence assessment

Aerodynamic Design: Front and rear wings, nose, diffuser and undertray, sidepods and bodywork, cooling ducts and radiator packaging; aerofoil selection, downforce and drag optimisation, aeromapping against ride height and speed; aero-vehicle dynamics integration, lap-time simulation (QSLs, Optimum Lap), Pacejka 6.1 tyre modelling

Test, Correlation & Instrumentation: On-car coast-down drag testing and tuft flow visualisation, telemetry and sensor instrumentation, logged data acquisition and analysis; CFD-to-test correlation; FMEA

CAD & Detailing: SolidWorks and Autodesk Fusion 360 (proficient); Siemens NX (learnt at Adani); CATIA V5 (surfacing course, 2026); AutoCAD; lofted wing geometry, surface modelling of nose cones and sidepods, parametric modelling, assembly design and packaging studies, 2D manufacturing drawings, GD&T

Structural Analysis & Optimisation: Abaqus, Ansys Mechanical and Ansys APDL, Fusion 360 FEA; static structural analysis, stress and deflection checks, topology and generative design optimisation, lightweight and composite structures, parameter-driven design optimisation

Materials & Manufacture: Carbon fibre composites — mould design, mould preparation and waxing, wet layup, resin infusion, prepreg, roll wrapping, vacuum bagging and finishing; lathe turning, CNC machining, water-jet cutting; aluminium 6061 and 7075-T6; outsourced manufacture with external suppliers; bill of materials and cost reporting

Programming & Software: Python (PyEuler2D — 2D Euler solver extended to rotating flow; post-processing and workflow automation), STAR-CCM+ Java macro automation, MATLAB, Linux command line, HPC and cluster job submission; Microsoft Word, Excel and PowerPoint

ENGINEERING EXPERIENCE

Imperial Formula Racing | Formula Student

Oct 2025 – Sep 2026

Simulations Engineer

London, UK

- Designed and developed the IFR-27 front and rear wing assemblies in SolidWorks and STAR-CCM+, balancing downforce against drag, packaging and the regulation aero boxes.
- Integrated the Pacejka 6.1 tyre model into the team's QSLs lap-time simulation, so an aerodynamic change could be judged in lap time rather than in coefficients on their own.
- Ran the aerodynamics and vehicle dynamics workstreams in parallel against a competition calendar that does not move.

Pravega Racing | Formula Student, SAE — VIT

Dec 2023 – Jan 2025

CFD Head

Vellore, India

- Owned the aerodynamic package on the PRV-25 across the electric and combustion cars — front wing, rear wing, nose, diffuser, undertray, sidepods and bodywork — designed in SolidWorks and Fusion 360 and developed in STAR-CCM+.

- Built aeromaps of downforce coefficient against ride height and speed for aero-vehicle dynamics integration, and correlated them against telemetry and the coast-down drag and tuft flow visualisation tests I ran on the car myself.
- Ran thermal and conjugate heat transfer CFD on the battery pack, motor and motor control unit through the team's transition to an electric platform, alongside radiator and cooling duct packaging.
- Automated the repeat work with STAR-CCM+ Java macros and Python post-processing, and wrote the procedures down so the same geometry gave the same answer regardless of who ran it.
- Carried designs through to hardware: FEA validation, moulds designed for the wings, nose and sidepods and laid up with the composites team, and 2D drawings with GD&T for aluminium parts outsourced to external suppliers.
- Taught newer members meshing methodology and the theory underneath it; handed the aerodynamics function to the next intake and mentored them through Formula Bharat 2025, where the team took 1st in Engineering Design, 1st in Cost and 3rd overall.

Pravega Racing | Formula Student, SAE — VIT

Mar 2022 – Dec 2023

Aerodynamics Engineer

Vellore, India

- Designed the complete aerodynamic package — front and rear wings, cooling ducts and bodywork — from concept CAD through CFD to manufactured parts.
- Ran aero-vehicle dynamics integration studies to quantify the aerodynamic contribution to braking, acceleration and cornering.
- Manufactured carbon fibre parts by hand with the composites team — mould preparation and waxing, wet layup, resin infusion, prepreg, roll wrapping, vacuum bagging and finishing — and turned the car's spacers on the lathe.
- Wrote and presented the technical design and cost reports to international judging panels at Formula Student Germany and Formula Student East, and led the cost report team at FS East 2023.

Adani Defence and Aerospace

Jan 2025 – Apr 2025

Project Trainee — Airframe and Aerodynamic Design

New Delhi, India

- Contributed to the airframe architecture and aerodynamic design of a loitering munition, from concept through to validation.
- Optimised wing profiles and planforms for lift-to-drag ratio, flight endurance and stability, and sized the control surfaces.
- Evaluated angle-of-attack behaviour and longitudinal and lateral stability margins using RANS CFD, and specified lightweight composite structures for strength-to-weight.

Tata Motors Passenger Vehicles Limited

Sep 2023 – Oct 2023

Project Trainee — EV Shop, Sanand Plant

Ahmedabad, India

- Worked on vehicle sensor integration, analysing logged sensor data to trace how sensor behaviour fed through into vehicle performance and efficiency, and ran an FMEA across the sensor subsystems.
- Applied Kaizen methodology on the production floor and designed a shadow board system for the workshop's torque tools.

RESEARCH & PROJECTS

Imperial College London, with Accuris (ESDU Group) — MSc Thesis

Feb 2026 – Sep 2026

"Derivation of the RANS Equations for Rotational Flow via Transformation Between Axes of Translation and Rotation"

- Derived the transformation of the 2D RANS equations from a stationary frame into a moving reference frame term by term, and rewrote the Spalart-Allmaras and k-omega turbulence models to sit inside that formulation for the unsteady flow around a rotating propeller.
- Extended PyEuler2D, a 2D Euler solver in Python, to rotating flow as an independent reference, and validated rotating NACA 0012 cases in Ansys Fluent against it — tracing the disagreement to domain sizing, rotating-frame setup and far-field boundary placement.

Vellore Institute of Technology — Undergraduate Final Year Project

Sep 2024 – Nov 2024

"Design and Development of an FSAE Rear Wing Mount using Generative Design and CFD"

- In a team of three, redesigned the rear wing mount with Fusion 360 generative design from CFD loads at the wing's centre of pressure, to a 2.5 safety factor and checked in static FEA: 45% lighter than the shape-optimised design (865.8 g to 477.1 g), with CFD showing the swan-neck layout gives 22.7% more downforce than a below-wing mount (128.4 N against 104.7 N).

ACHIEVEMENTS & AWARDS

- Formula Bharat 2025 — 1st in Engineering Design, 1st in Cost, 1st overall in Statics and 3rd overall in the Combustion category, with Pravega Racing (as mentor).
- Formula Student Germany 2023 — 18th overall and 9th in Engineering Design, Hockenheimring; Formula Student East 2023 — 12th in statics, Hungaroring.
- Formula Bharat 2023 — 2nd in Statics and 7th overall nationally; Autodesk Vortex 360 Designathon 2024 — 1st of 200 competing teams in the Design Challenge.
- VIT University Day 2023 — Achiever and Special Achiever awards; recognised by SAE-VIT for contribution as CFD Head of Pravega Racing.